

Online Medical Management System

A centralized platform connecting healthcare stakeholders. It offers secure, role-based access. Addresses healthcare management challenges.



Addressing Healthcare Challenges

1

Centralized Platform

Connecting key healthcare stakeholders - patients, doctors, labs, pharmacies, and hospitals.

2

Secure Access

Role-based access tailored to individual users.

3

Integrated Automation

Addresses challenges in healthcare management through integration and automation

Why Now?

1

Increasing Complexity

High intricacies of healthcare systems

2

Need for Efficiency

Faster, more efficient access to healthcare resources

3

Global Demand

For secure and scalable healthcare solutions

Problem Statement

Current Challenges

- Fragmented processes and lack of coordination among the roles and enterprises. 1.
- Inefficient appointment scheduling leads to longer wait times. 2.
- Limited access to consolidated medical records.3.
- Difficulty in processing and tracking insurance claims.4.

Impact

- Patients: Frustration due to delays and lack of transparency.
- Doctors: Limited access to patient medical histories affects diagnosis quality.
- Labs: Disconnected operations slow down report generation.
- Pharmacies: Stock mismanagement and prescription errors.

Proposed Solution

What Makes the Solution Unique?

- **Role-Based Access:** Ensures only authorized personnel can access specific data.
- **Centralized Management:** All healthcare operations are integrated into one platform.
- **Automation:** Reduces manual errors in scheduling, billing, and claim processing.
- **Global Accessibility:** Designed for diverse healthcare ecosystems.



Key Entities

1 Hospital Enterprise

2 Diagnostic Labs

3 Pharmacy Enterprise

4 Insurance Enterprise

Benefits and Impact

>> Faster Scheduling
Streamlined appointment booking.

>> Reduced Errors
Fewer billing mistakes.

>> Easier Access
To healthcare resources.

Functional Description

Login and Role-Based Access

Secure credentials for all users.
Role-specific dashboards to simplify navigation.

1

Patient Functions

Book and reschedule appointments.
Access the prescriptions, lab reports, and bills.

3

Pharmacy Functions

Verify prescriptions and track inventory levels.
Prevent stockouts through automated forecasting.

5

Doctor Functions

Access real-time patient data.
Update schedules and manage prescriptions.
Request diagnostic tests and access results.

2

Lab Functions

Review and approve test requests from doctors.
Conduct assigned tests and ensure accurate, timely results.
Communicate completed test results to doctors.

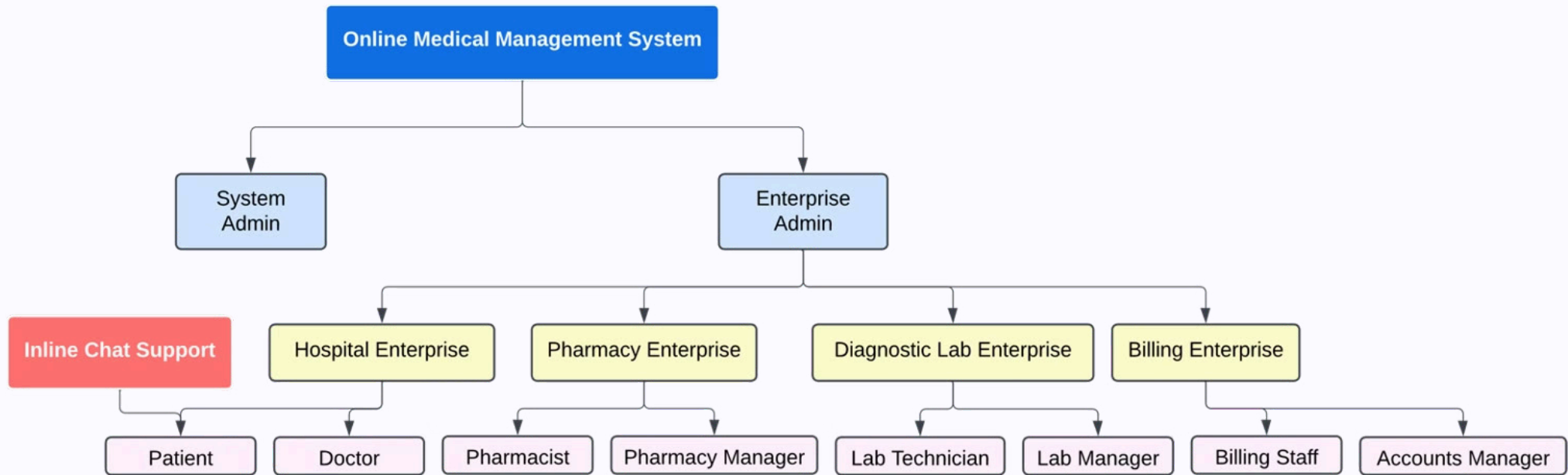
4

Insurance Functions:

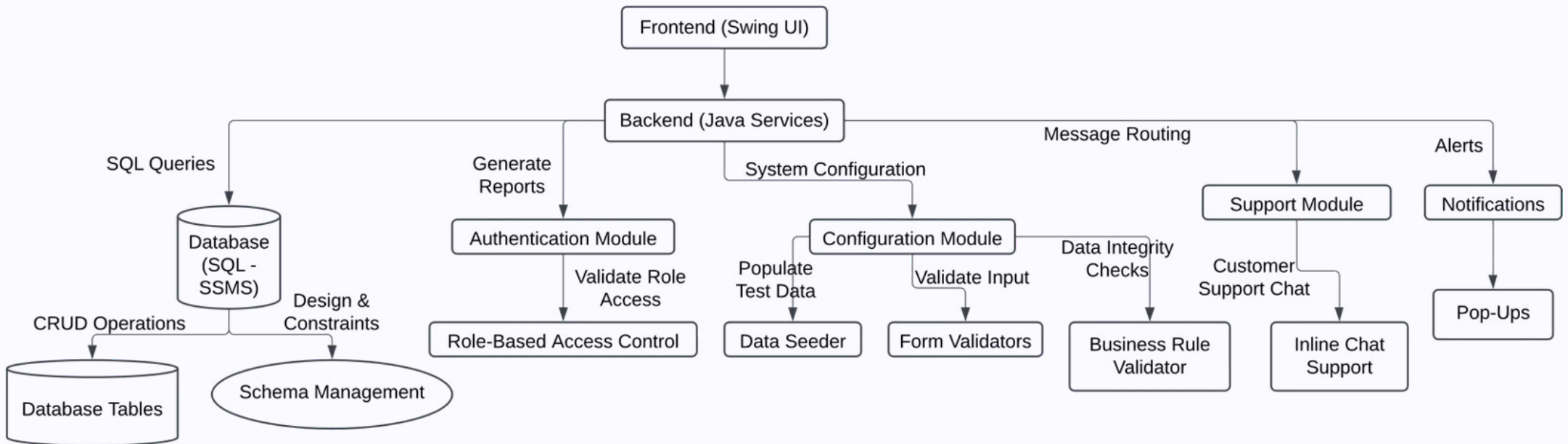
Validate insurance policies in real-time.
Process claims and share approval statuses with patients.

6

UML Diagram



High level Component Diagram



Key Use Cases

1

Appointment Scheduling

Doctors specify available time slots.

Patients book appointments online, reducing administrative load.

2

Lab Testing

Doctors request tests, and labs assign technicians based on workload.

Patients receive test updates via their dashboard.

3

Medication Management

Doctors send e-prescriptions directly to pharmacies.

Patients receive reminders for refills.

4

Billing and Insurance Claims

Integrated payment system supports credit cards, digital wallets, and insurance claims.

Automated claim approval reduces processing time.

5

Cross-Entity Collaboration

Doctors, labs, and pharmacies collaborate seamlessly to manage patient health.

Ecosystem Interaction

1

Patient Journey:

Registers → Books Appointment → Visits Doctor → Receives Prescription → Completes Lab Test → Accesses Medical History → Pays Bill.

2

Doctor-Lab Interaction:

Doctor schedules appointments → Doctor advises patient → Doctor orders test → Lab processes test request → Lab shares results → Doctor reviews.

3

Pharmacy Process:

Doctor prescribes medication → Pharmacy verifies prescription → Patient collects medication.

4

Insurance Workflow:

Patient undergoes treatment → Patient submits claim → Insurance Manager processes and approves/ declines the claim.



Benefits and Impact



Faster Scheduling

Streamlined appointment booking.



Reduced Errors

Fewer billing mistakes.



Efficient Access

Easier access to healthcare resources.



Transparency

Increased transparency in costs and processes.



Automation of Processes

Automation reduces overhead for hospitals.



Role-based Access Control

Role-based access ensures compliance with global health data standards like HIPAA.

Conclusion

Why This Matters:

- Bridging the gap between fragmented healthcare systems.
- Supporting scalable, adaptable solutions for global healthcare challenges.
- Emphasizing patient-centric innovation.

Future Vision:

- Expand features to include digital health and AI-driven diagnostics.
- Integrate wearable devices for real-time health monitoring.